

Sensyte

Project Proposal

Water Treatment Plant – PLC Data Monitoring & Motor Control System

Client: TechSync Solutions (Pvt) Ltd

Elevating Your Digital Presence

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Who we are

Sensyte (Pvt) Limited is a Sri Lanka based technology company formed through the merger of CircuitUp Technologies & Innvotics (pvt) Limited, delivering end-to-end engineering and product solutions across electronics, IoT, renewable energy, and electric vehicle (EV) industries.

We specialize in turnkey product development, covering electronic R&D, PCB and schematic design, embedded firmware and software development, IoT platforms, web dashboards, and OEM services. Beyond core electronics, Sensyte provides complete system solutions for renewable energy and EV-based products, from concept to deployment.

With strong expertise in multi-layer PCB design (up to 12 layers), power and high-speed electronics, ultra-low power devices, wireless communication, industrial systems, and energy and EV solutions, our multidisciplinary team delivers reliable, scalable, and future-ready technologies.

Sensyte serves clients across Australia, New Zealand, Singapore, the USA, and Sri Lanka, partnering closely to transform ideas into impactful, sustainable solutions.

Sensyte partners with innovators worldwide to turn complex ideas into reliable, production-ready solutions, delivering engineering excellence with global reach and real-world impact.

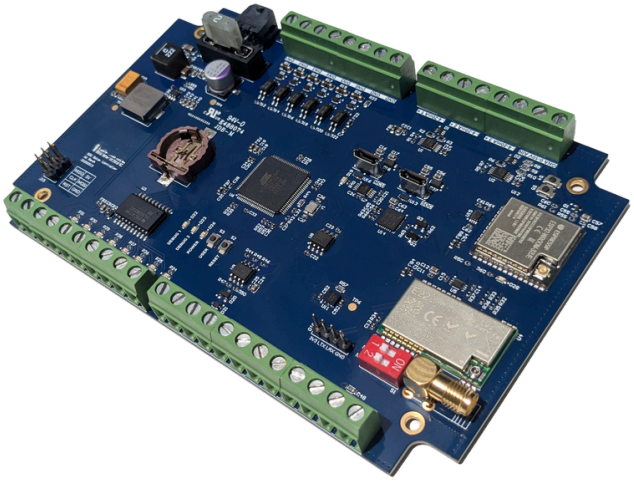
100+
Projects Completed

50+
Customers

NZ AUS SG USA
Overseas Projects

Our Services

Sensyte has experienced engineering to facilitate a variety of customer requirements. Including PCB design services, Firmware Development, Standalone software, Web Application, and Manufacturing services.



Electronic Product Development

Sensyte delivers complete electronic product development from concept and system design to hardware, firmware, and enclosure development. We design multilayer PCBs, develop reliable embedded firmware, and create production-ready enclosures, taking products from prototype to mass production. Our solutions support IoT, energy, wireless, and industrial applications, with a strong focus on performance, reliability, and manufacturability.



Mechanical Design

Sensyte offers comprehensive mechanical design and enclosure development services, including concept design, surface modeling, structural and thermal simulations, and design optimization for manufacturing. We support rapid prototyping through in-house 3D printing and coordinate injection molding for mass production. Using industry-leading tools such as SolidWorks, our team ensures mechanically robust, functional, and aesthetically refined designs ready for real-world deployment.

Our Services

Software Development

Sensyte provides full-stack software development to complement and scale electronic products. We develop secure and high-performance web dashboards, desktop applications, and mobile apps for both Android and iOS—enabling seamless device management, data visualization, and user interaction.

Our web solutions are built using modern technologies such as PHP, JavaScript, and Python, with frameworks like Laravel and Node.js, and integrated with cloud services, APIs, and SQL/NoSQL databases. We also develop Windows-based desktop applications using C# for device configuration, control, and specialized client workflows. From real-time monitoring to analytics-driven platforms, our software is designed to be reliable, scalable, and production-ready.



Proposed Solution

Overview

Sensyte proposes the design, development and deployment of an IoT-based PLC data acquisition and remote motor control system for the Client's Water Treatment Plant (WTP) network, covering five (5) locations.

At each location, a Sensyte Wi-Fi Slave Node (ESM30_V1) connects to the existing Programmable Logic Controller (PLC) over Ethernet using Modbus TCP and continuously reads the selected process data. The same node provides two (2) relay outputs that allow authorised operators to start and stop two (2) pump motors remotely.

The Slave Nodes publish their data to an MQTT broker. A Master Node in the control room subscribes to all five locations and keeps the data for the SCADA system. Option B also stores the data on a cloud server, so the plant can still be monitored and controlled if the SCADA system is down.

Objectives of the System

- Real-Time Process Monitoring
- Remote Motor Control
- Alarms & Notifications

Site Coverage – Five (5) Locations

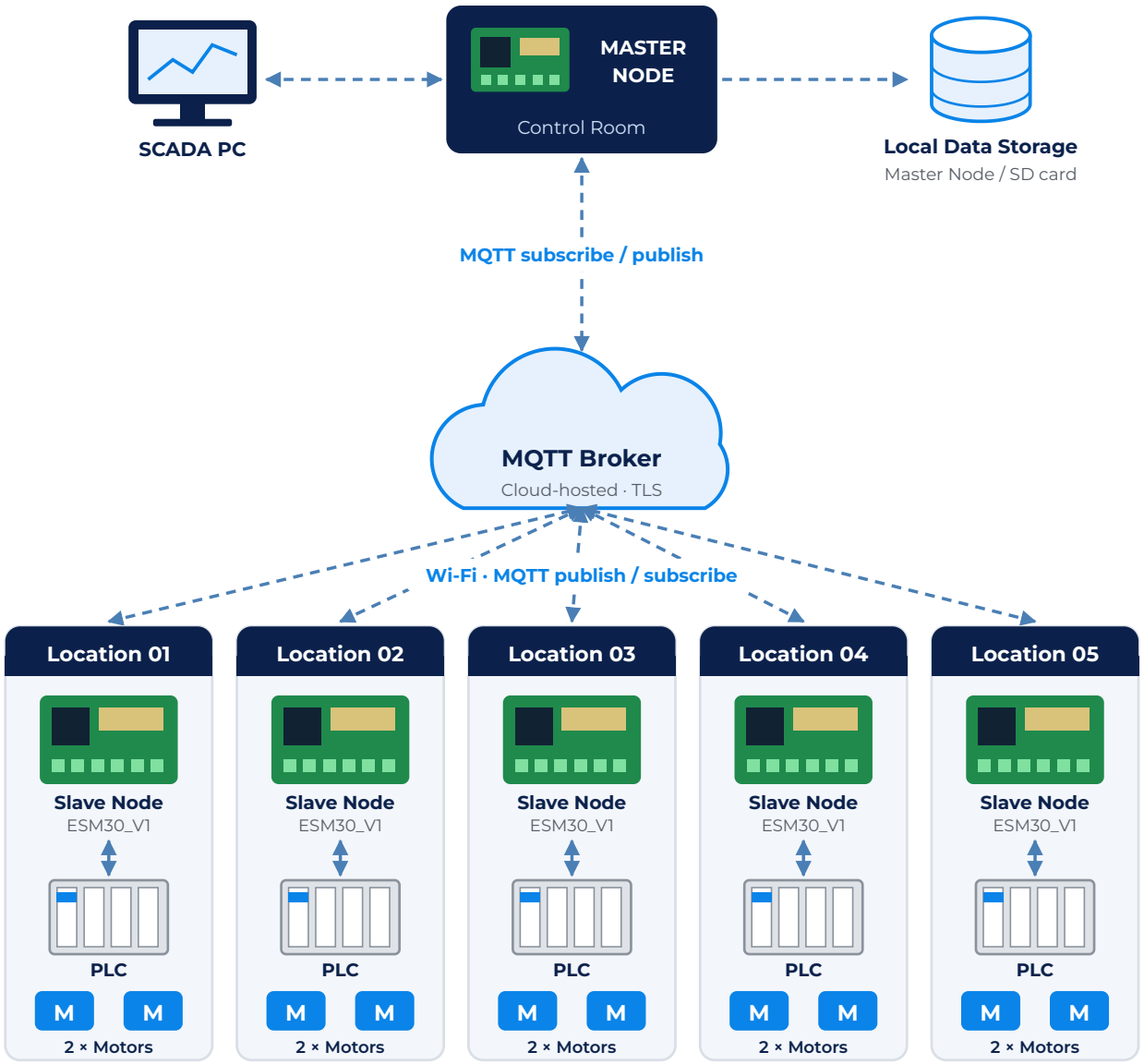
One Slave Node is installed in the control panel at each location, controlling two (2) motors. One Master Node is installed in the control room.

Location	Slave Node	Relay Outputs	Feedback Inputs
Location 01	1	2	2
Location 02	1	2	2
Location 03	1	2	2
Location 04	1	2	2
Location 05	1	2	2
Total	**5**	**10**	**10**

System Architecture

Option A – Master Node with Local Storage

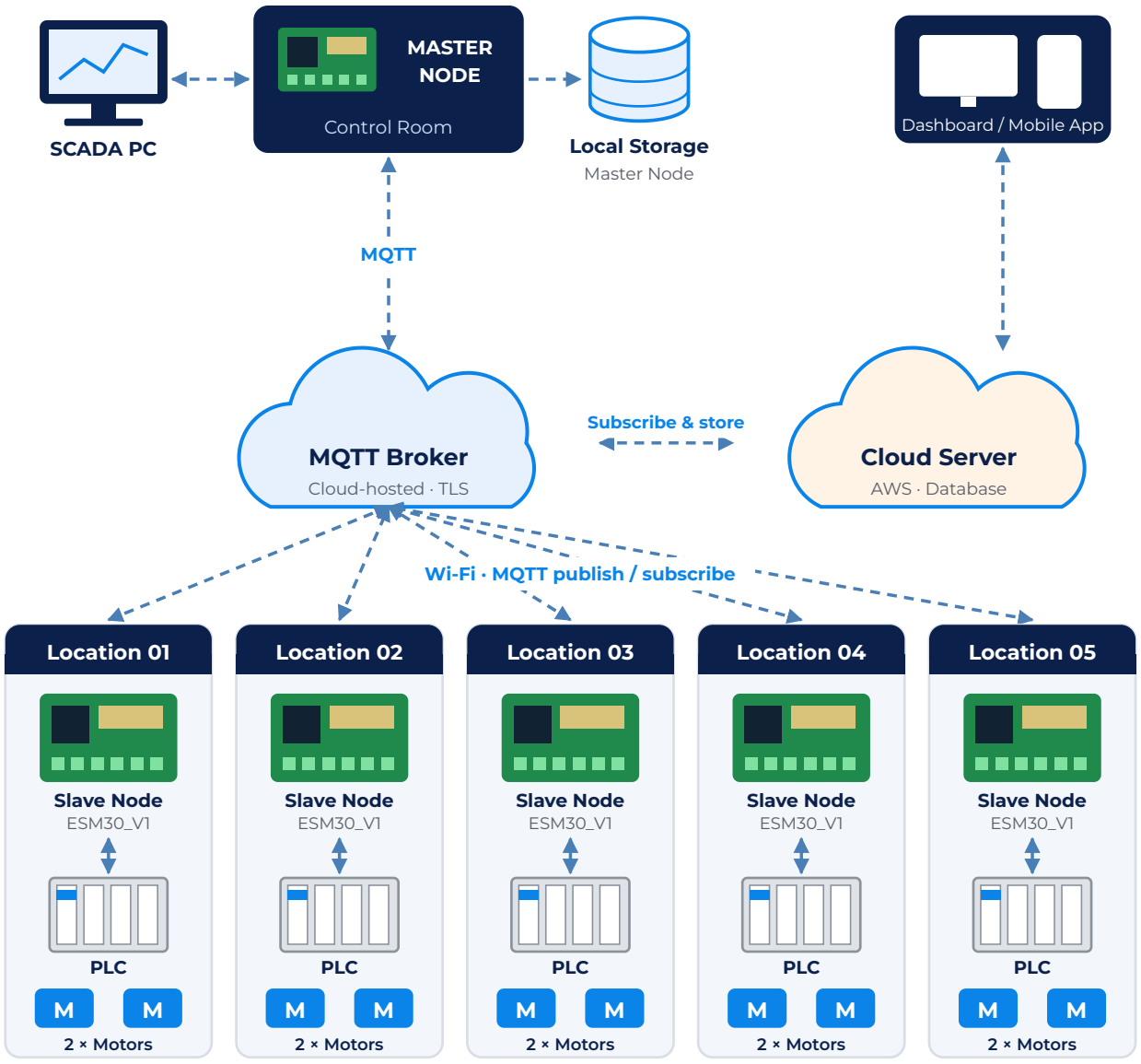
Each location has a Sensyte Slave Node (ESM30_V1) that reads the PLC over Modbus TCP and controls two (2) motors. The Slave Nodes publish their data to an MQTT broker over the site Wi-Fi. The Master Node in the control room subscribes to all five (5) Slave Nodes through the broker and keeps the data itself (no cloud database). Motor commands from the SCADA PC go back to the Slave Nodes through the same broker.



System Architecture

Option B – Master Node with Cloud Server

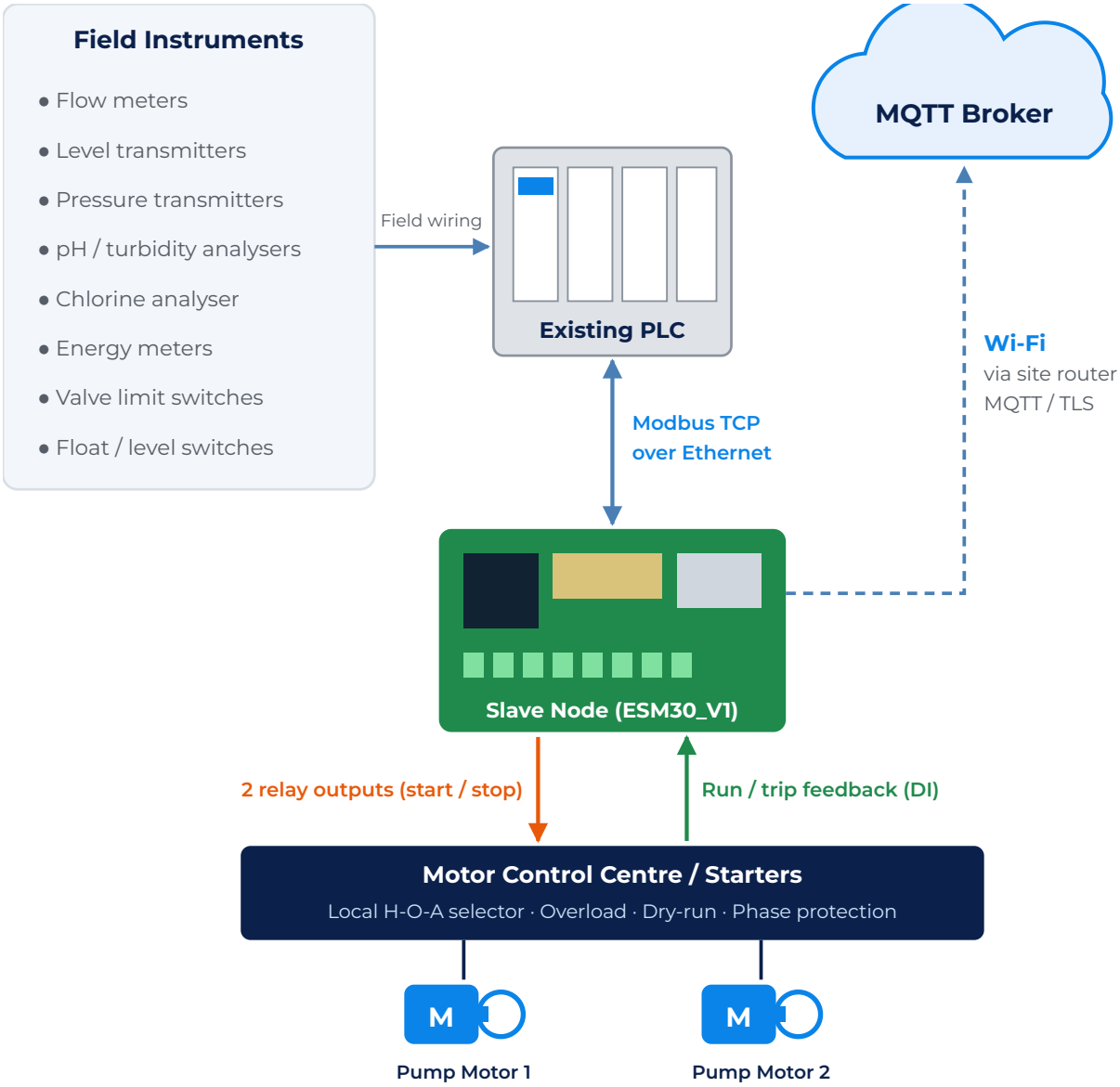
Option B uses the same Slave Nodes, MQTT broker and Master Node as Option A. In addition, a cloud server subscribes to the broker and stores all data in a cloud database. If the SCADA system is down, all five (5) locations can still be monitored and controlled from the web dashboard or mobile app.



System Architecture

Single Location – Detailed View

Field instruments remain connected to the existing PLC. The Slave Node reads PLC data using Modbus TCP over Ethernet and drives motor starters through its relay outputs, while auxiliary contacts return run / trip feedback to the Slave Node's digital inputs. Data and commands pass through the MQTT broker.



System Architecture

Slave Node Specification (ESM30_V1)

Feature	Specification
PLC Interface	1× Ethernet port – Modbus TCP (client)
Relay Outputs	2× dry-contact relays (one per motor), 5 A @ 250 VAC, with LED indication
Digital Inputs	2× opto-isolated inputs for motor run / trip feedback
Connectivity	Wi-Fi 2.4 GHz (IEEE 802.11 b/g/n) to the site router
Data Security	MQTT over TLS, device authentication
Data Buffering	Local storage with store-and-forward during outages
Configuration	BLE mobile app set-up, remote (OTA) firmware updates
Power & Mounting	24 VDC, surge protected, DIN-rail mount in control panel

Implementation Plan

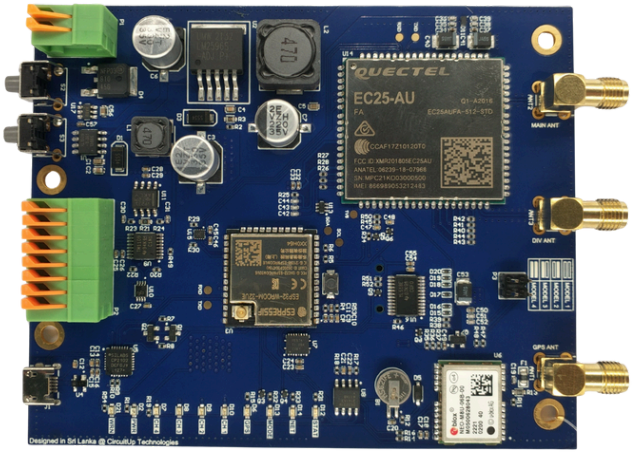
#	Phase	Timeline
1	Site survey and Modbus TCP register mapping (5 locations)	Week 1
2	Slave Node and Master Node configuration	Weeks 1 – 2
3	MQTT broker, SCADA integration and cloud dashboard (Option B)	Weeks 1 – 3
4	Testing with PLC / Modbus TCP simulator	Week 3
5	Installation and commissioning at all locations	Weeks 3 – 4
6	Site acceptance test, training and handover	Week 4

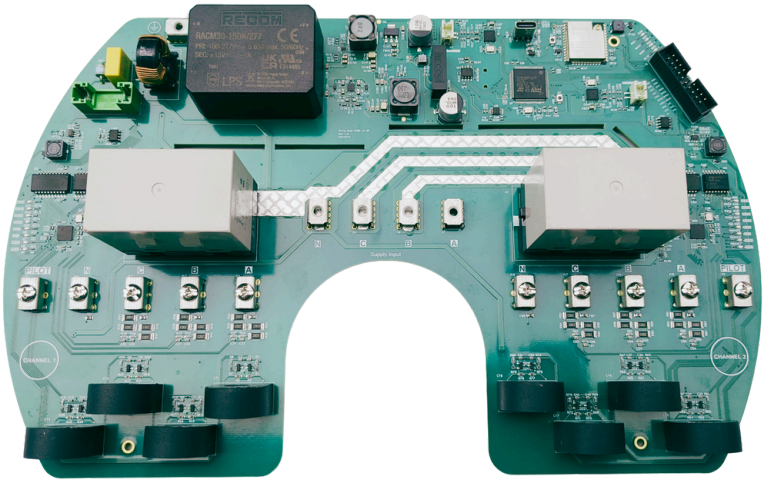
Project Portfolio

Hardware

Remote EV Battery Diagnostic Tool

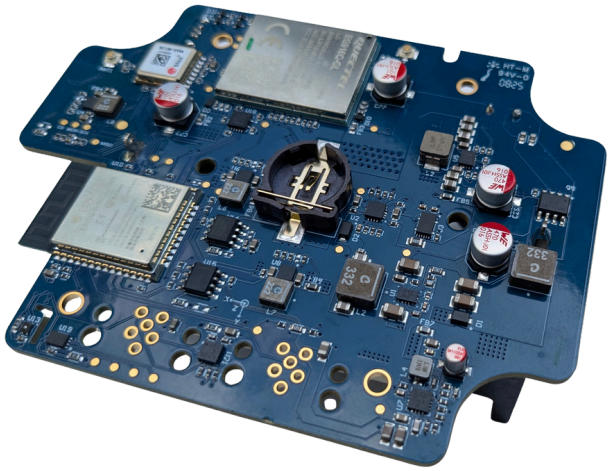
This project was designed as a remote EV battery health monitoring and diagnostic tool. CAN bus takes battery information and stores it in an SD card. Whenever the user requests, the logged file will be uploaded to the server using 4G.





EV Charger

An innovative dual-port Level 2 charger designed with the advanced power monitoring features and the wireless connectivity.

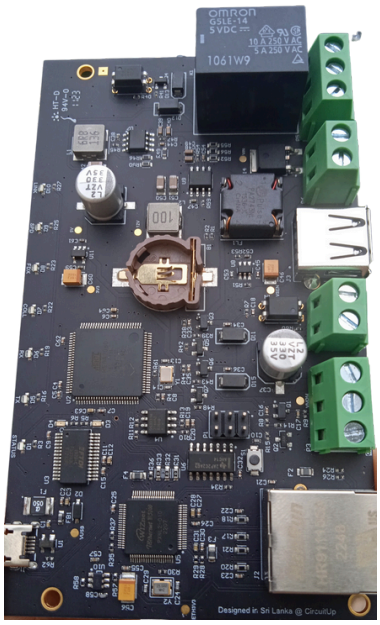


ULP Temperature Node

An ultra-low-power temperature device designed to operate for over one year on a single battery. It supports multiple sensors, FUOTA, and periodically sends data to the cloud over a 4G LTE network for reliable remote monitoring.

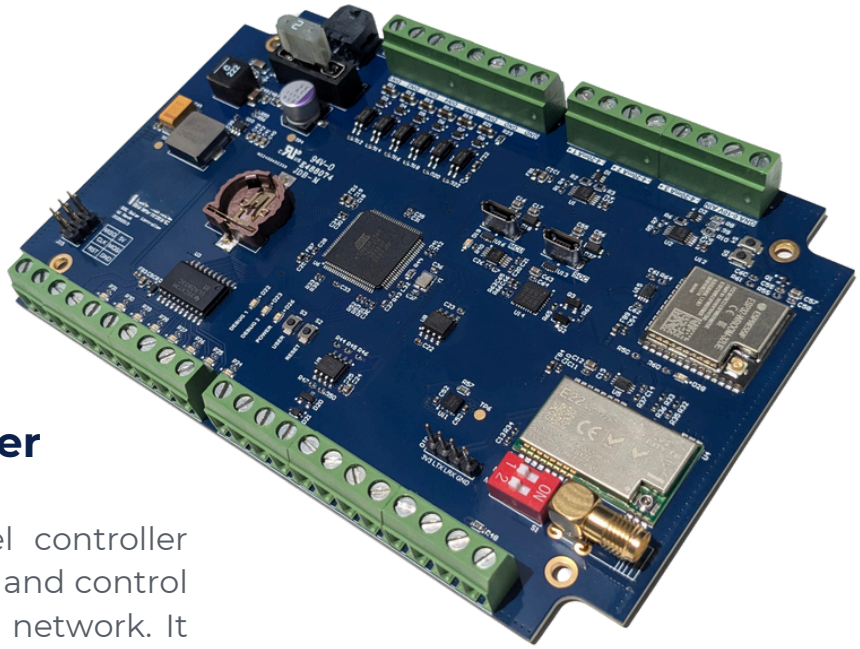
Project Portfolio

Hardware



Ethernet Sensor Controller

Designed to control specific sensors over the Ethernet. Also, multiple DC/DC convertors and power switches included.



Solar Panel Controller

An IoT-enabled solar panel controller enables remote monitoring and control of solar panels over a LoRa network. It supports multiple analog and digital inputs and outputs for robust operation, while built-in safety features ensure reliable and secure performance in industrial environments.

Project Portfolio

Mechanical



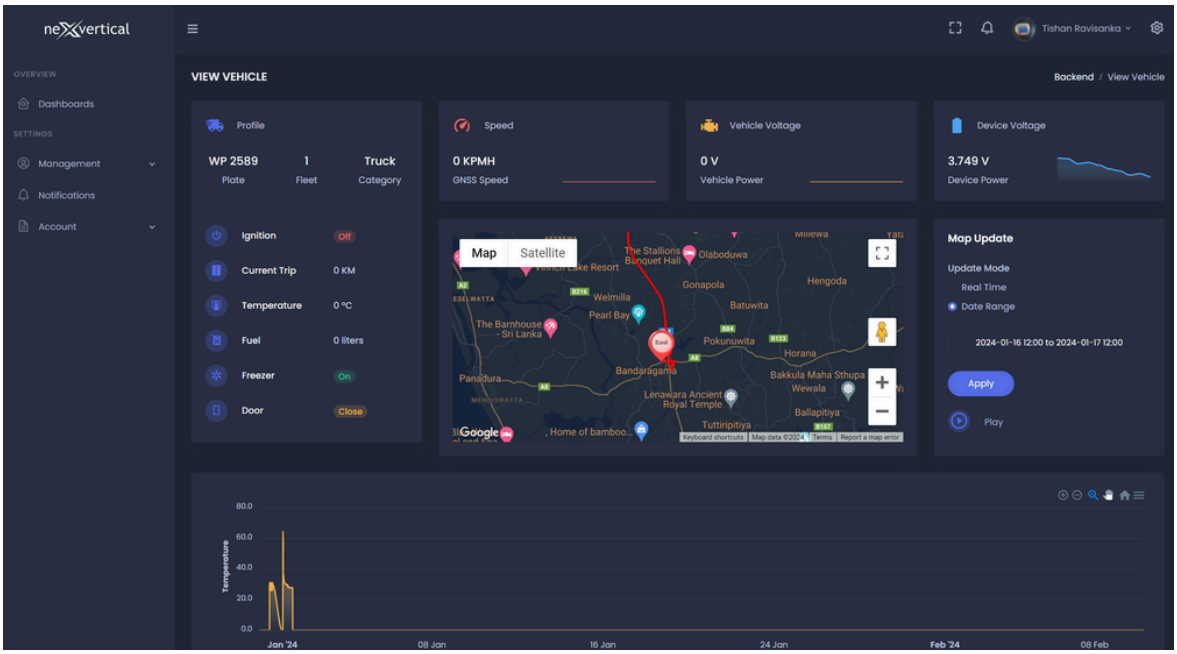
Enclosure Designs for LoRa WAN Temperature Node

Exclusively designed for a custom LoRaWAN PCB, this enclosure provides protection against dust and water splashes while offering two dedicated inlets for external connectors.

Project Portfolio

Software

Fleet Management System

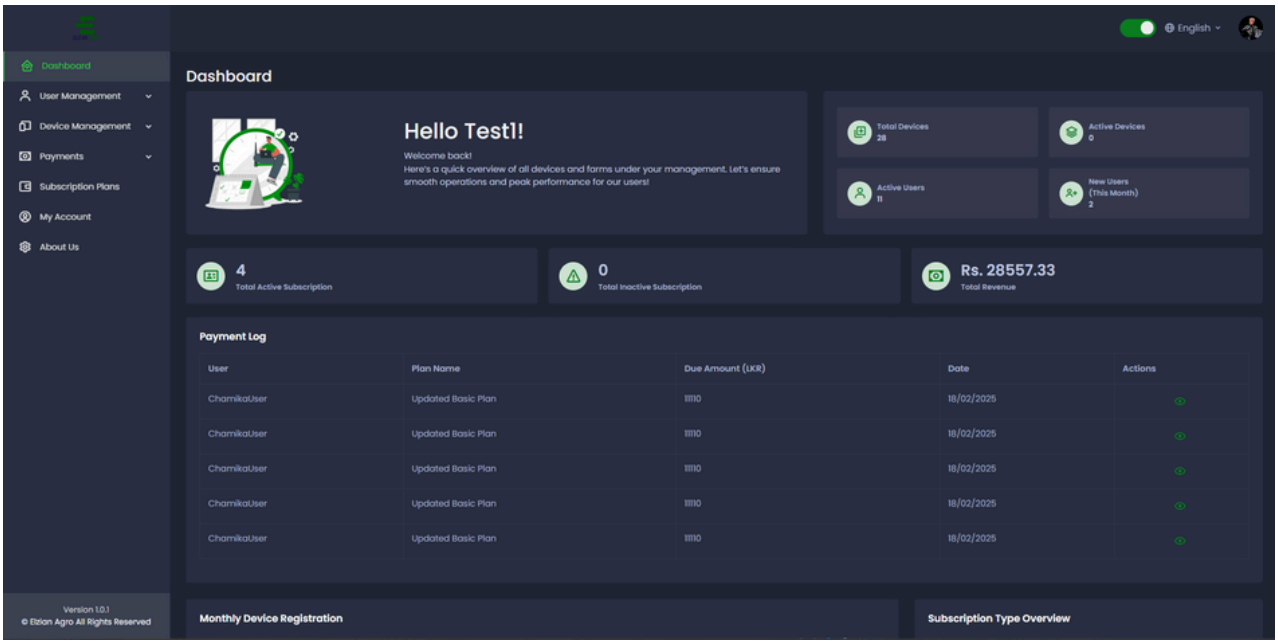


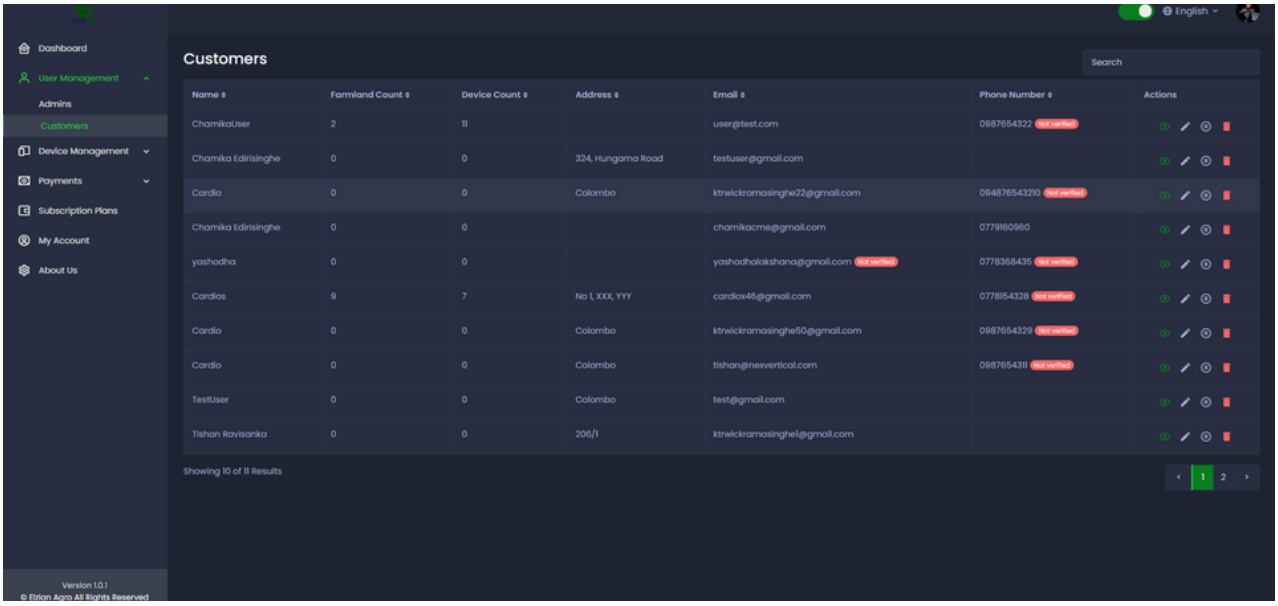


Project Portfolio

Software

IoT Platform for Smart Agriculture

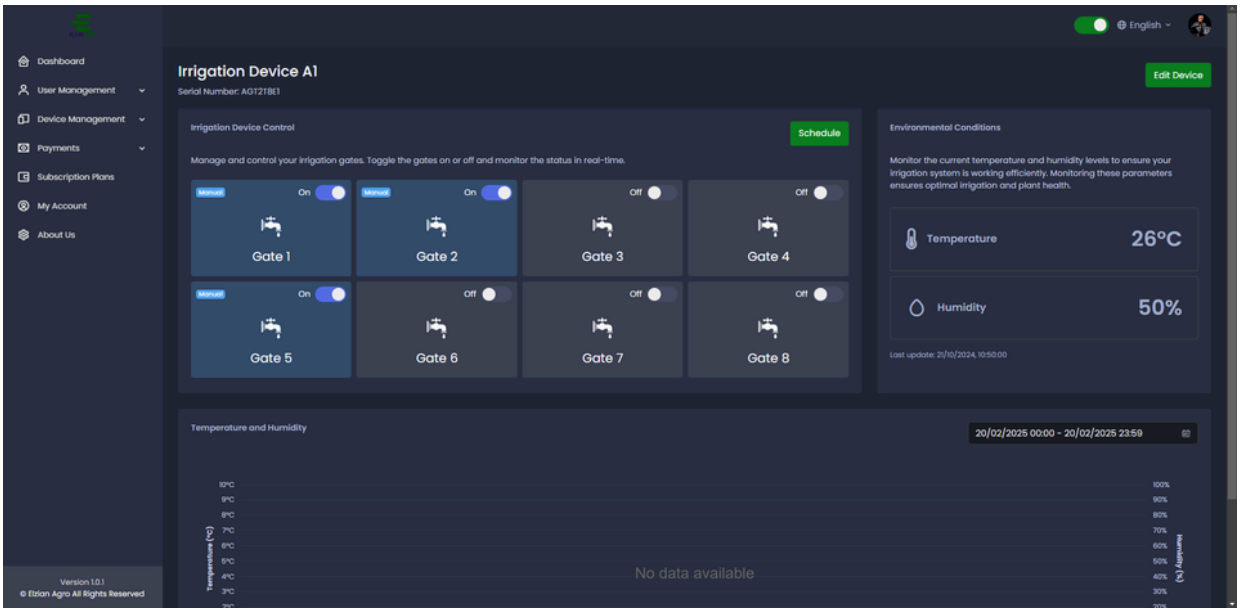


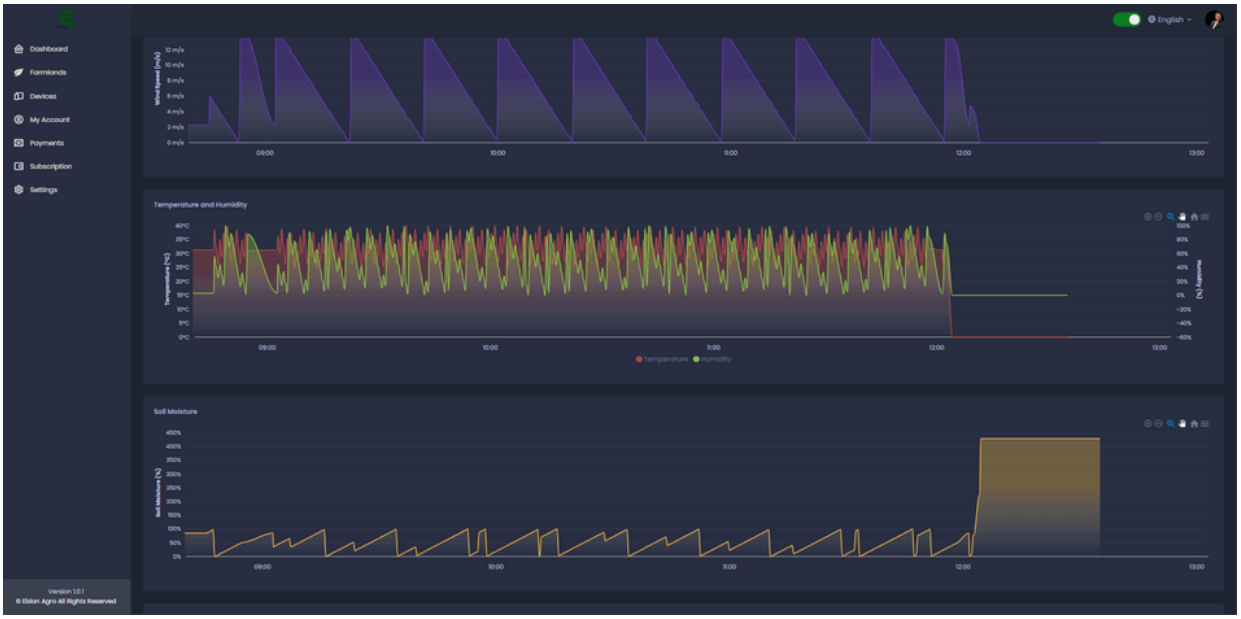


Project Portfolio

Software

IoT Platform for Smart Agriculture

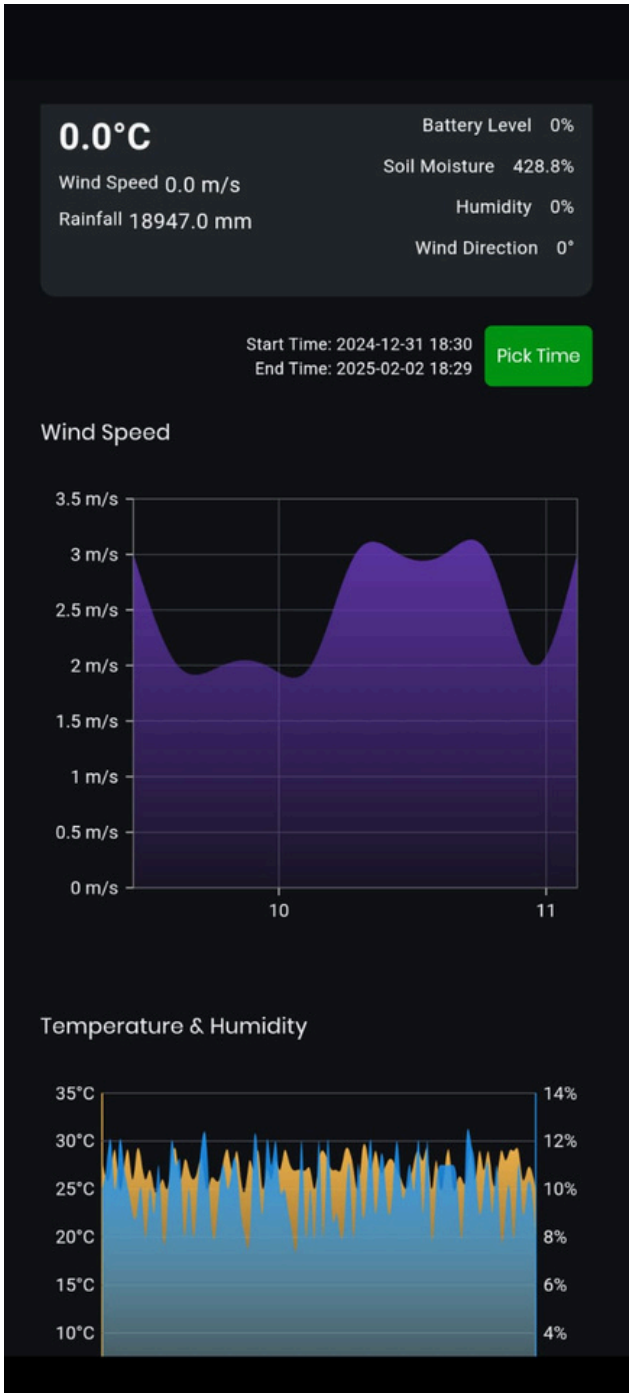


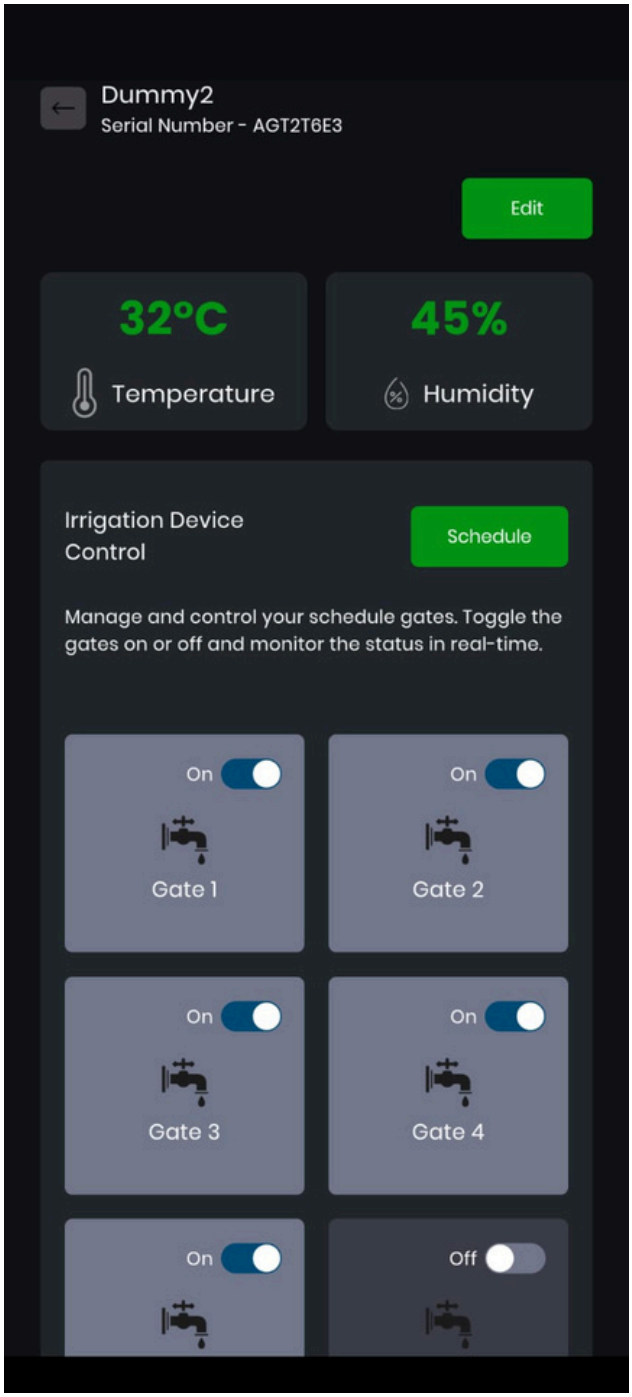


Project Portfolio

Software

IoT Platform for Smart Agriculture - Mobile App

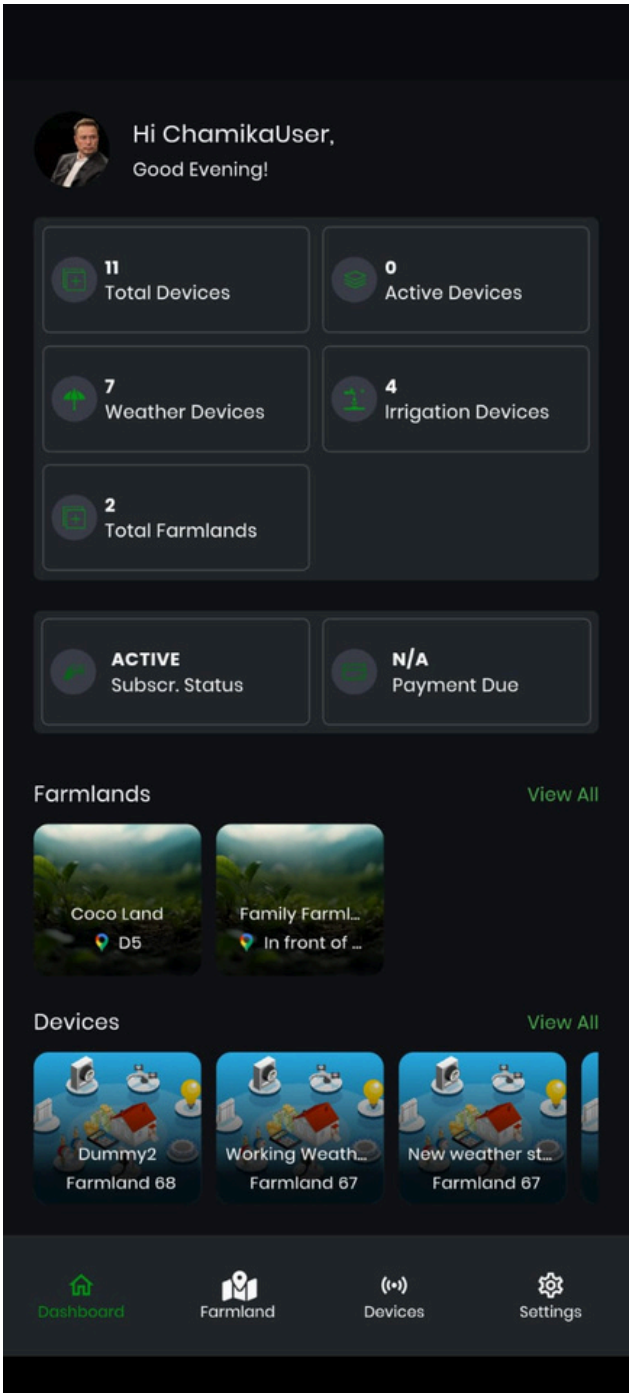


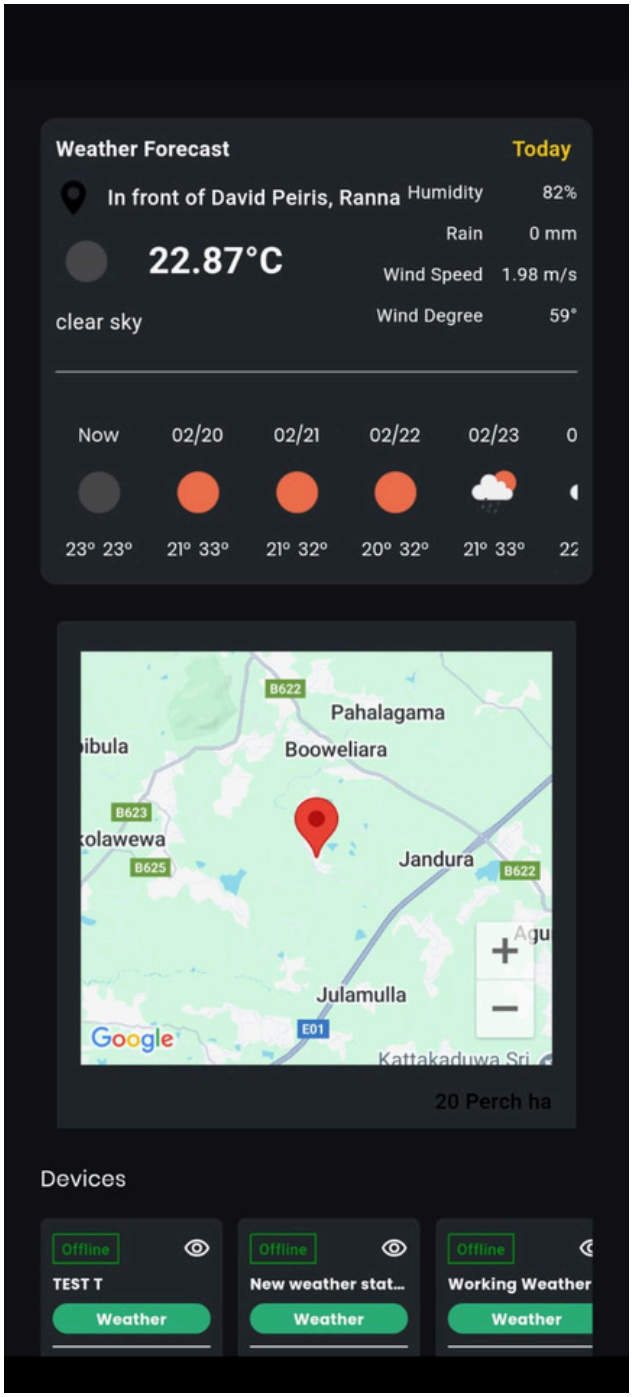


Project Portfolio

Software

IoT Platform for Smart Agriculture - Mobile App





Pricing

Two options are offered. Both use the ESM30_V1 Slave Nodes, the MQTT broker and the Master Node, with remote control of two (2) motors per location. Option B adds a cloud server that stores the data.

Option A – Master Node with Local Storage

#	Description	Amount (LKR)
1	Supply of Gateway ESM30_V1	330,000
2	Development cost	100,000
3	Cloud Services Charge (Annual Fee)	230,000
	One-time cost (items 1 + 2)	**430,000**
	Total for the first year (items 1 + 2 + 3)	**660,000**

> **Option A specification:** The Master Node subscribes to the Slave Nodes via the MQTT broker and keeps the data itself (SCADA PC / SD card). No cloud database.

Option B – Master Node with Cloud Server

#	Description	Amount (LKR)
1	Supply of Gateway ESM30_V1	330,000
2	Development cost	300,000
3	Cloud Services Charge (Annual Fee)	345,000
	One-time cost (items 1 + 2)	**630,000**
	Total for the first year (items 1 + 2 + 3)	**975,000**

> **Option B specification:** All data is also stored on the cloud server. If the SCADA system is down for any reason, the motors can still be monitored and controlled through the cloud dashboard and mobile app.

All prices are in Sri Lankan Rupees (LKR) and exclude applicable taxes. After the first year, only the annual Cloud Services Charge (item 3) applies.

Cloud Subscription

What the Subscription Includes

Service	Option A	Option B
Cloud-hosted MQTT broker with secure (TLS) connection	Yes	Yes
Data storage in the AWS cloud database	—	Yes
Cloud data rate of one (1) second interval	—	Yes
Data backups and data archives	—	Yes
Web dashboard and mobile app access	—	Yes
Monitoring and control when SCADA is down	—	Yes
Platform hosting, maintenance and security updates	Yes	Yes

Annual Fee

Option	Annual Fee (LKR)
Option A – Master Node with Local Storage	230,000
Option B – Master Node with Cloud Server	345,000

Payment Terms for Cloud Subscription

- The annual fee is payable in full, in advance, for each subscription year.
- The first subscription year starts on the date of commissioning and handover.
- The renewal invoice will be issued 30 days before the end of each year, and is payable within 14 days.
- If the renewal is not paid within 30 days after expiry, cloud access may be suspended. Stored data will be kept for 90 days during suspension.
- A Wi-Fi connection with internet access must be provided by the Client at each location.

Payment & Conditions

General Conditions

- The estimated project duration shall be up to four (4) weeks from the advance payment and receipt of the PLC information.
- A detailed and final project timeline shall be provided after the final technical and commercial discussion and mutual agreement.
- The scope of deliverables shall include the following:
  - **Devices** – Supply of five (5) Sensyte ESM30_V1 Slave Nodes, one (1) per location, each with two (2) relay outputs to control two (2) motors, and one (1) Master Node for the control room, including configuration and commissioning.
  - **PLC Integration** – Reading of the agreed data points from the PLC at each location. The Client shall provide the PLC make / model, Modbus TCP register map, Ethernet access to the PLC, and access to the panels. Changes to the existing PLC program shall be done by the Client or PLC vendor, or quoted separately.
  - **Mobile Application** – Monitoring, control and device configuration app delivered in APK (Android) format.
  - **Backend & Dashboard (Option B)** – Deployment on an AWS server. Hosting is charged under the annual Cloud Services Charge (see Pricing).
  - **Source Code & Project Files** – Not included in this agreement. May be made available upon additional payment, under a separate written agreement.
- Additional devices or locations may be supplied upon the Client's request, at prices confirmed in writing.
- Electrical wiring of relay outputs to motor starters / MCC panels shall be done by the Client's licensed electrician under Sensyte's guidance, or under a separate contract.
- A stable Wi-Fi network with internet access at each location shall be provided by the Client.

Payment & Conditions

Payment Conditions

- The Client shall pay 50% of the total project fee as an advance prior to the commencement of any work.
- The balance 50% of the total project fee shall be settled within four (4) weeks from the date on which the project deliverables are formally handed over to the Client.

warranty

- Warranty Period
  - A six (6) month warranty shall be provided from the date of sale.
- Warranty Coverage
  - The warranty covers manufacturing defects only.
  - The warranty does not cover physical damage or damage resulting from external factors, including but not limited to:
    - High-voltage spikes or electrical surges
    - Lightning or thunder strikes
    - Fire, water, or environmental damage
    - Improper use, mishandling, or unauthorized modifications
- Dashboard Bug Fixes
  - Any software bugs identified within six (6) months from the date of deployment in the dashboard system shall be fixed free of charge.
  - This applies only to defects related to the originally delivered functionality and scope.
- Dashboard Maintenance
  - Ongoing maintenance, updates, and support for the dashboard system may be provided under a separate annual maintenance contract, subject to mutually agreed terms and pricing.

Contact Us

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